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Optimization of dilute acid and hot water pretreatment of different lignocellulosic biomass: A comparative study

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ABSTRACT

Pretreatment of biomass to alter their recalcitrant structures is an essential step to obtain high yield of products via bioconversion processes. In this study, main emphasis was to compare the results evaluated in terms of total reducing sugars (TRS) yield after acid and hot water pre-treatment process performed with laboratory scale equipment using different lignocellulosic biomass. The biomass chosen for this purpose i.e. sugarcane bagasse and bamboo were collected from Guwahati, Assam and their physico-chemical characteristics were examined using X-ray diffraction (XRD), Fourier transform infrared (FTIR) spectroscopy and thermo gravimetric analysis (TGA) including proximate and ultimate analysis. Crystalinity of the biomass used was observed to be 33.15% and 31.29% for sugarcane bagasse and bamboo respectively. Hot water and dilute acid pretreatment allows selective solubility of hemicellulose which improves the accessibility of enzymes for cellulose hydrolysis. The highest yield of TRS was observed at run order 8 for both acid and hot water pretreatment (23.49 and 26.50 gL^{-1}) with respect to sugarcane bagasse. But, the pretreatment results obtained for bamboo was slightly different to that of sugarcane bagasse. The highest yield of TRS was obtained at run order 8 for acid (15.6 gL^{-1}) and run order 10 for hot water (17.98 gL^{-1}) pretreatment respectively. Irrespective of biomass type, hot water pretreatment process produced more TRS than acid pretreatment process.

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1. Introduction

The fluctuation of crude oil price and rising concern on environment, renewable energy resources have become top priority for many countries. The upsurge of biofuel production paves the way to search for lignocellulosic biomass materials. Unlike fossils and nuclear fuels, availability of lignocellulose materials is not limited around the world. Almost all countries are abundant with the lignocellulosic resources depending on their geographical locations. The policy in sustainability of renewable energy is the main motivation for inflating shares

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on renewable energy. About 118 countries, of which more than half are developing nations, currently have fixed renewable energy goals and 109 nations have guidelines in support of renewable energy in the power sectors. The usage of bioenergy may increase up to 1791 million tons oil equivalent within 2020 [1]. Lignocellulosic biomass is generated from green plants, which converts sunlight into energy stored in the form of carbohydrate molecules such as sugars through a photosynthesis process, and includes all land and water-based vegetation, as well as all organic wastes generated via various human activities. Therefore, the biomass can be defined as an organic matter in which the energy of sunlight is stored in chemical bonds [2]. Characterization of biomass is necessary before using it as feedstock in a bio-refinery for the production of bio-fuel and other valuable green chemicals. The compositions of biomass within the same species may be altered with varying seasons and climatic conditions. The relative quantities of principle building block elements of lignocellulosic material, i.e. cellulose, hemicellulose and lignin content are often required in studies involving fractionation [3], biofuel production [4] or synthesis of green chemicals [5]. High Lignin content reduces the thermo-chemical yield and bio-energy product. Therefore, pretreatment is an essential step to disrupt lignin structure and improve the bio-fuel yield for the fermentation process of lignocellulosic biomass. The utilization of both cellulose and hemicellulosic sugars like hexose, pentose, etc. present in a typical biomass hydrolysate is essential for the economical production of biofuels. Lignocellulosic material is very resistant to enzymatic breakdown; hence pretreatment is required in order to enhance the susceptibility of biomass to enzymes. The main goal of all pretreatment methods is to create a platform for optimizing yield of fermentable sugars from biomass, in order to increase biofuel production invariably. However, the final sugar yield depends not only on biomass characteristics, but also on their interaction with pretreatment conditions and enzyme formulations. The most successful physico-chemical pretreatments include thermo-chemical treatments such as steam explosion, grind milling, hot water/auto hydrolysis, acid treatment and currently hydrothermal hydrolysis is being investigated [6].

In this paper, main emphasis was on to compare the results evaluated in terms of TRS yield of acid and hot water pretreatment process performed with laboratory scale equipment using different lignocellulosic biomass. The biomass chosen for this purpose are bamboo and sugarcane bagasse. India has the richest bamboo reserves in the world [7]. The states of North-East India is known for its wide natural resources, among various types of biomasses bamboo and sugarcane forms an significant section in the traditional landscape of North East India [8]. The total production of sugarcane in Assam is 1.06 million tons, whereas in totality India produces 292.30 million tons [9]. The usage of bamboo as a fuel wood in India is 1145 tons [10]. Initially the selected biomass samples were characterized for their physico-chemical composition. Physical properties were estimated by proximate analysis, XRD and TGA. Chemical properties were determined by FTIR, CHNS-O and Soxhlet extractions followed by sugar contents analysis. The results of this study are useful to design the technology that is suitable for more efficient production of bio-fuel and for process economic analysis.

2. Materials and methods

2.1. Materials

The biomass used in this study i.e. bamboo (*Bambusa cacharensis*) and sugarcane bagasse were collected from Guwahati, Assam. After harvesting, biomass samples were sun dried in an open air then chopped and sieved (mesh size - BSS 30) to get homogeneous powder (diameter of particles lower than 0.5 mm). The powdered biomass samples were used as the starting material for characterization. AR grade sulfuric acid (98% Emplura) and Dinitrosalicylic acid were obtained from Merck India Ltd. and HiMedia respectively.

2.2. Characterization of biomass

2.2.1. CHNSO analysis

The carbon, hydrogen, nitrogen, sulphur and oxygen are the fundamental elements of any biomass which are commonly known as CHNSO. For the analysis purpose biomass sample (1.0 mg) was placed in a tin boat assortment to determine the percentage composition of carbon, hydrogen, sulphur and nitrogen. The percentage of oxygen was determined by means of difference and the results are tabulated in Table 1.

2.2.2. Ash content

The materials of ash are classified into three groups such as water soluble (free ionic form) organic solvent soluble and precipitated. The inorganic materials loosely adhered with the biomass known as extractable ash content. These are easily separated from biomass by simple extraction or washing techniques. Extractable ash is the result of soil remaining in the biomass. Structural ash materials are adhered with biomass strongly. The cations (K^+, Mg^{2+}, Ca^{2+}, Al^{3+}, Mn^{2+}, Fe^{3+}, Cu^{2+}, and Zn^{2+}) present in the ash has a positive inhibition effect on enzyme, which helps to hydrolyze the cellulosic substrates into glucose and other readily fermentable sugars [10]. The estimation of ash present in the biomass is necessary before processing the biomass for biofuel production. The procedure to estimate the amount of ash present in biomass was adopted from the NREL protocol. The biomass of 0.2(±0.02) gm was placed in an oven dried moisture free crucibles and heated up to 575 ± 25 °C in a muffle furnace for three hours. The dry weights of ash content in biomass samples were calculated using Equations (1) and (2).

$$\text{Dry wight of the sample} = \frac{\text{weight(air dried sample)} \times \text{Total solid(\%)}}{100} \quad (1)$$

Table 1 – Proximate and ultimate analysis of biomass samples.

Name of the biomass	Proximate analysis				Ultimate analysis					
	Ash (%)	Volatile matter (%)	Fixed carbon (%)	Moisture (%)	Carbon (%)	Hydrogen (%)	Nitrogen (%)	Oxygen (%)	Sulfur (%)	Calorific Value ($MJKg^{-1}$)
Bagasse	1.7 ± 0.04	78.44 ± 0.2	13.62 ± 0.3	6.2 ± 0.6	42.1 ± 0.7	5.2 ± 0.4	0.43 ± 0.2	52.5 ± 0.6	0.06 ± 0.1	17.20
Bamboo	1.62 ± 0.03	68.15 ± 0.1	23.92 ± 0.2	7.2 ± 0.5	42.8 ± 0.5	6.3 ± 0.3	0.35 ± 0.3	50.65 ± 0.8	Nil	17.97

$$\text{Ash}(\%) = \frac{\text{weight(ash + crucible)} - \text{weight(crucible)}}{\text{dry weight of the sample}} \times 100 \quad (2)$$

2.2.3. *Calorific value determination*

The calorific value of biomass was determined using a static bomb calorimeter; sealed parr 1108 according to the protocol reported by Naik et al. (2010) [11]. The biomass sample pellets approximately weighing 1 g each was put in contact with the platinum wire attached with cotton thread, which was placed on the ignition port. Approximately 1 ml of distilled water was added to the bomb and filled with 20 bar oxygen at 25 °C. In the isothermal jacket, calorimeter was placed with the air gap separation of 10 mm between all surfaces. The bomb calorimeter was submerged in a cane filled with distilled water, and the calorimeter jacket was maintained at a constant temperature by circulating water at 25 °C. The discharged ignition energy from about 40 V platinum wires was evaluated using change in potential across a 1256 or 2900 μF capacitor.

2.2.4. *XRD analysis*

The crystalline structure of cellulose was first established by Carl Von Nägeli in 1858 [12]. Crystalline regions are formed due to the hydrogen bonds between cellulose chains and Van der Waals forces between the glucose molecules in cellulose and hemicelluloses [13]. Cellulose crystallinity was obtained from wide angle X-ray diffractometer (WAXD) (Brucker D8 Advanced X-ray diffraction measurement system). The samples with particle size less than 120 μm were transferred into the glass sample holder and analyzed under plateau conditions. The radiation was generated with an accelerated voltage of 40 KV and current of 40 mA. The scan scope was between 7° and 40° with a step size of 0.02°, determination time of 1 s per 0.02° and constant rotation speed of 20 rpm. Crystallinity index along with the intensity of the main crystalline plane (002) and the amorphous fraction was calculated using the intensity at~18°of 2θ for amorphous fraction and maximum intensity of (002) plane at ~22.8° of 2θ for crystalline fraction [14]. The crystallinity of biomass samples was determined using Equation (3) as stated below.

$$\text{CRI} = \frac{(I_{002} - I_{amr})}{I_{002}} \times 100 \quad (3)$$

CRI = Crystallinity Index.
I_{002} = Intensity at 22.8°
I_{amr} = Intensity at 18°

2.2.5. *FTIR analysis*

FTIR spectroscopy was used for accurate prediction of chemical composition of biomass (Shimadzu, IR Affinity 1). Dried potassium bromide (KBr) and biomass sample were mixed thoroughly (sample and KBr weight ratio of 1:100). The mixed powder was scanned and recorded between 4000 and 400 cm^{-1} with a resolution of 4 cm^{-1}. The reproducibility of spectra of all the samples was verified on three sample preparations; the scan of 64–100 was used with a resolution of 4 cm^{-1}. Baseline and other necessary corrections for penetration depth and frequency variations were applied using Shimadzu IR solution 1.5 software supplied with the equipment.

2.2.6. *Thermo gravimetric analysis (TGA)*

TGA is a rapid method to estimate the temperature assisted decomposition profile of material and kinetics of its thermal decomposition. In the present study, thermogravimetry analysis of two different biomass samples was carried out to understand the thermal decomposition pattern of materials. The prepared biomass sample was stored in air tight plastic containers until they were needed for the TG analysis. Sample amount used for the analysis was almost the same 10 ± 0.5 mg. The non-isothermal analysis was performed in thermo gravimetric analyzer (model TG/SDTA851e, METTLER TOLEDO, Switzerland) at the constant heating rate of 10 °$Cmin^{-1}$ under nitrogen atmosphere. The continuous records of weight loss vs temperature data was obtained and analyzed using inbuilt software (STAR).

2.3. *Response surface methodology*

The classical approach does not depict the combined effect of all the process parameters. It is also time consuming and requires number of experiments to determine the optimum levels, which may be unreliable thereby elevating the overall process cost. These limitations of the classical method can be eliminated by optimizing all the process parameters collectively by statistical experimental design such as Response Surface Methodology (RSM). Hence, the combined effects of various parameters on the TRS yield by acid hydrolysis from biomass viz. temperature, biomass loading (B.L.) and treatment time were carried out using Central Composite Design (CCD) of RSM by MINITAB. Further, a numerical optimization methodology was employed to optimize the TRS yield. RSM is a collection of mathematical and statistical techniques that is useful for developing, improving and optimizing the processes and to evaluate the relative significance of several process parameters even in the presence of complex reaction conditions. The main objective of RSM was to estimate the optimum

Table 2 – Experimental range and levels of independent process variables.

Factor	Name	Low	High
(a) For acid pretreatment			
A	Temperature (°C)	80	100
B	Biomass loading (wt%)	5	10
C	Treatment Time (min)	10	20
(b) For hot water pretreatment			
A	Temperature (°C)	100	150
B	Biomass loading (wt%)	5	10
C	Treatment Time (min)	10	20

operational conditions of the process. This methodology is widely used in chemical engineering and applied sciences to optimize process variables. RSM offers certain advantages like higher percentage yield, reduced process variability, closer confirmation of output response to nominal and target achievement.

Design of Experiment (DOE) is a pre-planned approach for finding cause and effect of relationships between the factor variables. The purpose of statistically designing an experiment is to collect common relationship between various factors affecting the process towards finding the most suitable conditions. It is essential that an experimental design methodology be economical for extracting the maximum amount of complex information, a significant reduction in experimental time, saving both material and cost. CCD, Box-Behnken and Doehlert designs are among the principal response surface methodologies used in experimental design. CCD is the most widely used technique in the experimental design which is suitable for fitting a quadratic surface and helps to optimize the effective parameters with a minimum number of experiments and also to analyze the interaction between the parameters.

2.4. *Building empirical model*

The first step in RSM is to find out suitable approximation for the true functional relationship between dependent variable and the set of independent variables i.e. single-response model using RSM corresponding to independent variables. Then a mathematical model in the form of a second-order polynomial is framed to predict the response as a function of independent variables involving their interactions. Generally, the system behavior is explained by the following quadratic Equation (4).

$$Y = \beta_0 + \sum_{i=1}^{k} \beta_i x_i + \sum_{i=1}^{k} \beta_{ii} x_i^2 + \sum_{i=1}^{k} \sum_{j=i+1}^{k} \beta_{ij} x_i x_j + \varepsilon \quad (4)$$

Where Y is response (dependent variable), β_o is constant coefficient, β_i, β_{ii}, β_{ij} are coefficients for the linear, quadratic and interaction effect respectively, x_i, x_j are factors (independent variables), and error is represented by ε.

In this work, a second order polynomial equation was obtained using the uncoded independent variables as mentioned in Equation (5).

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_{11} x_{11} + \beta_{22} x_{22} + \beta_{33} x_{33} + \beta_{44} x_{44} + \beta_{12} x_1 x_2 + \beta_{13} x_1 x_3 + \beta_{14} x_1 x_4 + \beta_{23} x_2 x_3 + \beta_{24} x_2 x_4 + \beta_{34} x_3 x_4 \quad (5)$$

Multiple regression analysis technique was used to evaluate the coefficient of the model. Three factors A, B, and C was studied and their low and high levels are given in Table 2a and b. The TRS yield was studied with a standard RSM design namely CCD. Twenty experiments were conducted in duplicate according to the scheme mentioned in Table 3. Design Expert Version 7.1.6 (Stat Ease, USA) software was used for

Table 3 – Design matrices and TRS yield of dilute acid pretreatment and hot water pretreatment of sugarcane bagasse and bamboo.

Run order	Coded factors			TRS (gL^{-1})			
				Acid pretreatment		Hot water pretreatment	
	A	B	C	Bagasse	Bamboo	Bagasse	Bamboo
1	−1	−1	−1	8.06	5.05	9.02	7.81
2	1	−1	−1	12.27	7.83	14.21	9.21
3	−1	1	−1	11.83	8.27	12.05	10.25
4	1	1	−1	17.61	10.18	20.14	12.56
5	−1	−1	1	11.48	9.52	11.59	11.21
6	1	−1	1	17.38	12.51	19.84	14.92
7	−1	1	1	16.01	13.45	17.08	16.14
8	1	1	1	23.49	15.6	26.51	16.84
9	−1.68	0	0	10.61	10.34	11	12.02
10	1.68	0	0	20.61	14.46	22.08	17.98
11	0	−1.68	0	11.44	6.49	11.51	8.45
12	0	1.68	0	19.77	11.8	21.29	13.51
13	0	0	−1.68	9.47	5.44	10.28	8.14
14	0	0	1.68	17.29	13.76	19.44	16.02
15	0	0	0	16.25	11.99	17.91	14.86
16	0	0	0	16.28	12	18.11	14.78
17	0	0	0	16.22	11.98	17.85	14.62
18	0	0	0	16.25	11.99	17.95	14.8
19	0	0	0	16.26	11.98	18	14.91
20	0	0	0	16.24	12	17.88	14.98

Table 4 – Extractives (g) from 100 g of biomass samples.

Biomass	Hexane extract (wt%)	Ethanol extract (wt%)	Water extract (wt%)	Cellulose	Hemicellulose	Lignin
Bagasse	0.9 ± 0.3	6.5 ± 0.3	7.7 ± 0.4	40.15 ± 0.3	22.78 ± 0.2	20.81 ± 0.2
Bamboo	0.5 ± 0.2	7.5 ± 0.4	6.6 ± 0.2	38.29 ± 0.43	25.51 ± 0.1	20.17 ± 0.3

regression and graphical analysis of obtained data. Optimum values of the selected variables were obtained by solving the regression equation and by analyzing the response surface contour plots. The variability of dependent variables was explained by multiple coefficients of determination, R^2 and the model equation were used to predict the optimum values and subsequently to elucidate the interaction between the factors within the specified range.

2.5. *Acid and hot water pretreatment procedure*

Both acid (dilute sulfuric acid concentration was maintained at 0.5% wv^{-1}) and hot water pretreatment process were carried out in a 200 ml reactor at varying time (10–20 min), biomass loading (5-10 wt%) and temperature interval (80 °C-150 °C) as per the RSM process conditions mentioned above. After reaction, liquid and solid parts were analyzed accordingly. The supernatant liquid was analyzed for TRS estimation using 3, 5-Dinitrosalicylic acid (DNS) method as adopted by Miller, 1959 [15].

3. Results and discussion

3.1. *Characterization of biomass*

The details of basic physical properties and chemical compositions of the biomasses i.e. sugarcane bagasse and bamboo are shown in Table 1. The weight percent of ash and moisture content of both the biomass was almost similar and lies between the ranges of 1.6%–1.7% and 7.2%–6.2% for bamboo and sugarcane bagasse respectively. The fixed carbon content in bamboo was only 1.75 times higher than sugarcane bagasse. The ultimate analysis revealed that oxygen content in bagasse was slightly higher than bamboo. The more amount of oxygen in the bagasse was responsible for higher volatile matter content in the biomass sample. Calorific value which represents the important property of biomass for its energy content was found to be almost similar for both the biomass. Among the studied biomass, bagasse shown lower sulfur content, whereas in case of bamboo it was below the detectable limits. The lower sulfur content in the biomass implies less SO_x emission during the gasification which paves the way to use this biomass as a potential feedstock for gasification as well.

The yield of hexane, ethanol and water extracts for different biomass sample is shown in Table 4. Insignificant difference was observed in terms of total extracts. Hexane and

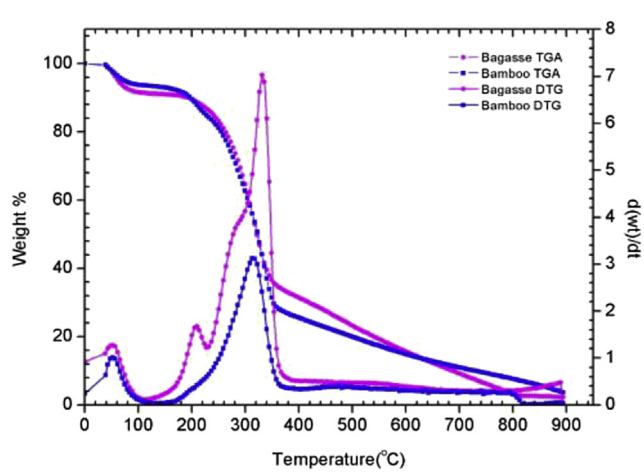

Fig. 1 – Thermo gravimetric and differential thermo gravimetric analysis of biomass samples.

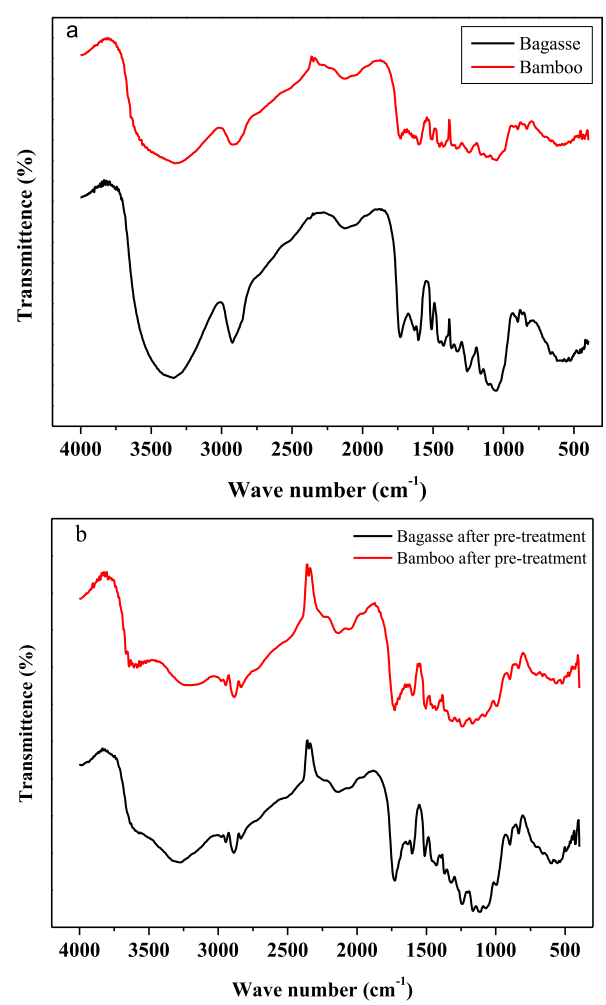

Fig. 2 – FTIR analysis of biomass samples (a) before pre-treatment; (b) after pre-treatment.

Table 5 – Coefficients and *t*-values for hot water pretreatment of (a) Bamboo (b) Sugarcane bagasse using central composite design.

Term	Coef	SE Coef	"t" value	Probability
(a) Bamboo				
Constant	14.82	0.17	88.80	0
A	1.27	0.11	11.48	0
B	1.63	0.11	14.74	0
C	2.47	0.11	22.27	0
A*A	−0.11	0.11	−0.99	0.34
B*B	−1.33	0.11	−12.31	0
C*C	−0.94	0.11	−8.70	0
A*B	−0.12	0.14	−0.83	0.43
A*C	0.23	0.14	1.59	0.14
B*C	0.28	0.14	1.90	0.09
b) Sugarcane bagasse				
Constant	17.93	0.21	83.44	0
a	3.63	0.14	25.46	0
b	2.75	0.14	19.29	0
c	2.56	0.14	17.97	0
a*a	−0.39	0.14	−2.84	0.02
b*b	−0.44	0.14	−3.20	0.01
c*c	−0.99	0.14	−7.12	0
a*b	0.51	0.19	2.74	0.02
a*c	0.55	0.19	2.95	0.01
b*c	0.4	0.19	2.15	0.06

ethanol extraction process was performed to remove the non-polar elements. It was observed that bagasse contained more amounts of non-polar materials than bamboo. The final raffinate after the solvent extraction process was extracted with water. The purpose of this step was to extract polar materials present in the residual biomass sample. Water extractives were relatively more for bamboo compared to bagasse. Both the biomass samples showed good amount of sugar content and crystallinity of the lignocellulosic biomasses were found to be 33.15% and 31.29% for sugarcane bagasse and bamboo respectively. The weight loss profile of biomass sample was evaluated using the constant isothermal ramp program in DTG (Fig. 1). Almost similar devolatilization pattern was noticed for both the biomass samples, onset temperature of devolatilization was in the range of 200–220 °C which corresponds to 5% weight loss with respect to final weight loss. The maximum weight loss occurred at 310–360 °C for both the biomass followed by slow and continuous weight loss. First step corresponds to the primary devolatilization, whereas the later was attributed to the degradation of heavier chemical structures in the solid matrix. The weight loss profile in case of both the biomass can be explained as follows. The major decomposition of bagasse and bamboo was noticed at around 335 °C and 320 °C respectively, which corresponds to the decomposition of structural sugars components. A long tail of devolatilization was observed in both the biomass samples which may be due to the high percentage of mannose or lignin in its composition (Table 4). FTIR spectra of lignocellulosic material were recorded on a Shimazdu spectrophotometer in the range of 400–4000 cm^{-1} with a resolution of 4 cm^{-1}. The FTIR spectra of these lignocellulosic materials are shown in Fig. 2. Analysis of these spectra was primarily based on formerly published data [16]. It was observed that spectrum signals at around 1000 cm^{-1} to 1250 cm^{-1} exhibited in the spectra were originated from hydroxyl and ether groups of polysaccharides. The bands at 1512 and 1464 cm^{-1} relate to aromatic ring stretching to authorize the aromatic ring structure of lignin molecules. In addition, a peak at 1731 cm^{-1} was found in BRM, and hemicellulose is probably related to the acetyl and/or uronic acid groups of hemicelluloses or the ester linkages of ferulic and *p*-coumaric acids with lignin. All correlations were developed by means of least squares regression analysis and the regression coefficients of the correlations ranges from 0.85 to 0.9.

Table 6 – Coefficients and *t*-values for dilute acid pretreatment of (a) Bamboo (b) Sugarcane bagasse using central composite design.

Term	Coef	SE Coef	"t" value	Probability
(a) Bamboo				
Constant	11.99	0.01	2283.34	0
X	1.23	0.003	352.24	0
Y	1.58	0.003	452.32	0
Z	2.47	0.003	709.21	0
X*X	0.15	0.003	43.81	0
Y*Y	−1.002	0.003	−295.53	0
Z*Z	−0.84	0.003	−248.09	0
X*Y	−0.21	0.005	−46.96	0
X*Z	0.06	0.005	12.36	0
Y*Z	0.18	0.005	39.82	0
(b) Sugarcane bagasse				
Constant	16.25	0.01	1160.36	0
x	2.94	0.01	316.70	0
y	2.47	0.01	266.04	0
z	2.32	0.01	250.14	0
x*x	−0.23	0.01	−25.36	0
y*y	−0.23	0.01	−25.56	0
z*z	−1.02	0.01	−112.52	0
x*y	0.39	0.01	32.43	0
x*z	0.42	0.01	34.90	0
y*z	0.19	0.01	15.75	0

3.2. *Pretreatment process optimization*

Based on preliminary studies, biomass loading, temperature and time of operation were identified as the major factors affecting the TRS yield of dilute acid and hot water pretreatment processes (Table 2a and b). In the present work, these variables were statistically optimized with the help of 3^3 factor

Table 7 – Analysis of Variance (ANOVA) for the response surface quadratic model for total reducing sugars (TRS) by dilute acid pretreatment.

Source	DF	Seq SS		F-value	
		Bagasse	Bamboo	Bagasse	Bamboo
Regression	9	293.83	162.44	27688.79	108892.81
Linear	3	275.49	137.85	77880.45	277213.27
Square	3	15.38	23.94	4346.48	48150.66
Interaction	3	2.97	0.65	839.46	1314.49
Residual error	10	0.012	0.002		
Lack-of-fit	5	0.01	0.001	4.90	3.14
Pure error	5	0.002	0.000		
Total	19	293.84	162.45		

Table 8 – Analysis of Variance (ANOVA) for the response surface quadratic model for total reducing sugars (TRS) by hot water pretreatment.

Source	DF	Seq SS		F-value	
		Bagasse	Bamboo	Bagasse	Bamboo
Regression	9	395.79	177.60	158.33	117.84
Linear	3	373.17	141.49	447.83	281.65
Square	3	16.84	34.97	20.21	69.61
Interaction	3	5.78	1.14	6.94	2.28
Residual error	10	2.78	1.68		
Lack-of-fit	5	2.73	1.60	61.28	20.70
Pure error	5	0.045	0.08		
Total	19	398.57	179.27		

design using RSM. The experimental design matrix and results are shown in Table 3. The highest yield of TRS was observed at run order 8 for both acid and hot water pretreatment with sugarcane bagasse. The yield obtained for both the pretreatment process was 23.49 and 26.51 gL^{-1} respectively. The pretreatment results obtained for bamboo was slightly different to that of sugarcane bagasse. The highest yield of TRS was obtained at run order 8 for acid (i.e. 15.6 gL^{-1}) and run order 10 (i.e. 17.98 gL^{-1}) for hot water pretreatments respectively. Irrespective of the biomass type hot water pretreatment process produced more sugars than the acid pretreatment process. At higher temperature, pH of the water was reduced drastically. This study used a Student's 't' test to observe any differences between the analyzed and calculated data

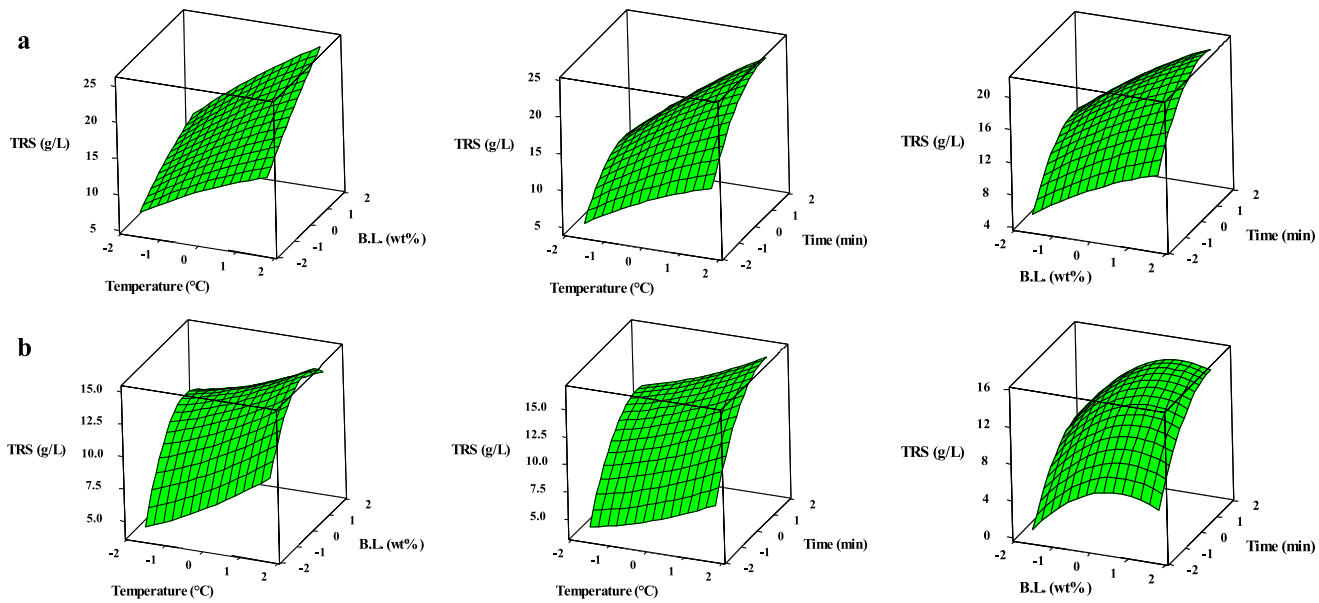

Fig. 3 – 3D response surface plots of TRS yield for dilute sulfuric acid pre-treatment (a) sugarcane bagasse, (b) bamboo.

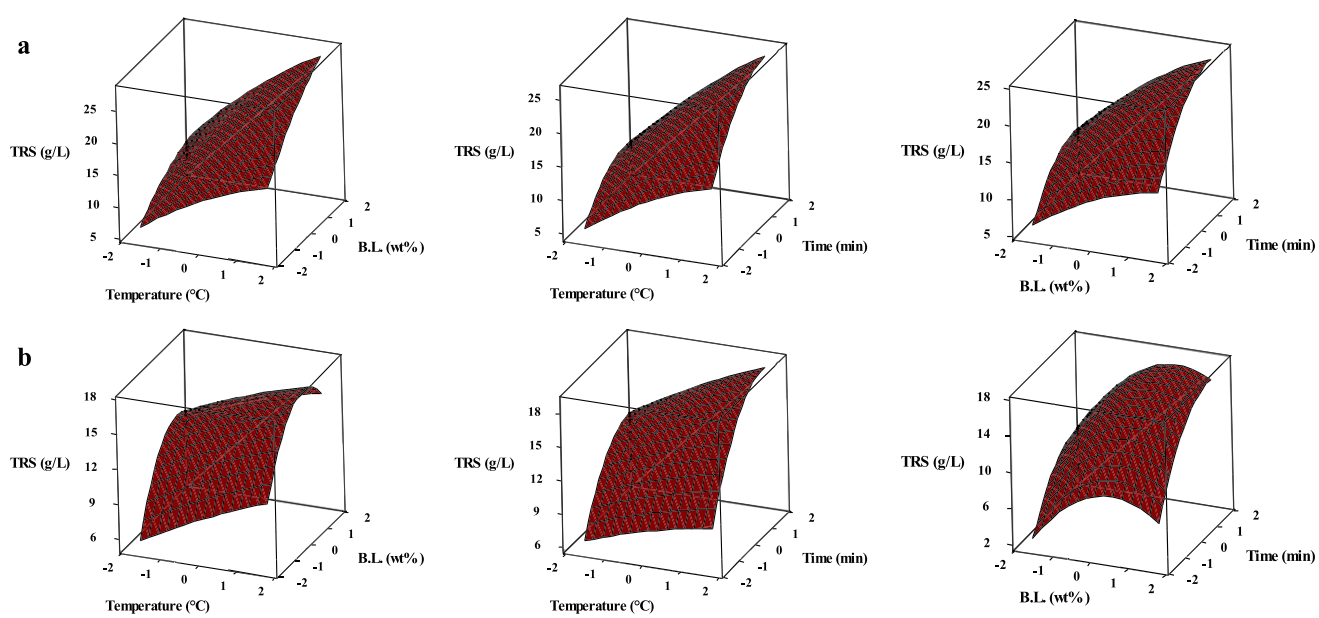

Fig. 4 – 3D response surface plots of TRS yield for hot water pre-treatment (a) sugarcane bagasse, (b) bamboo.

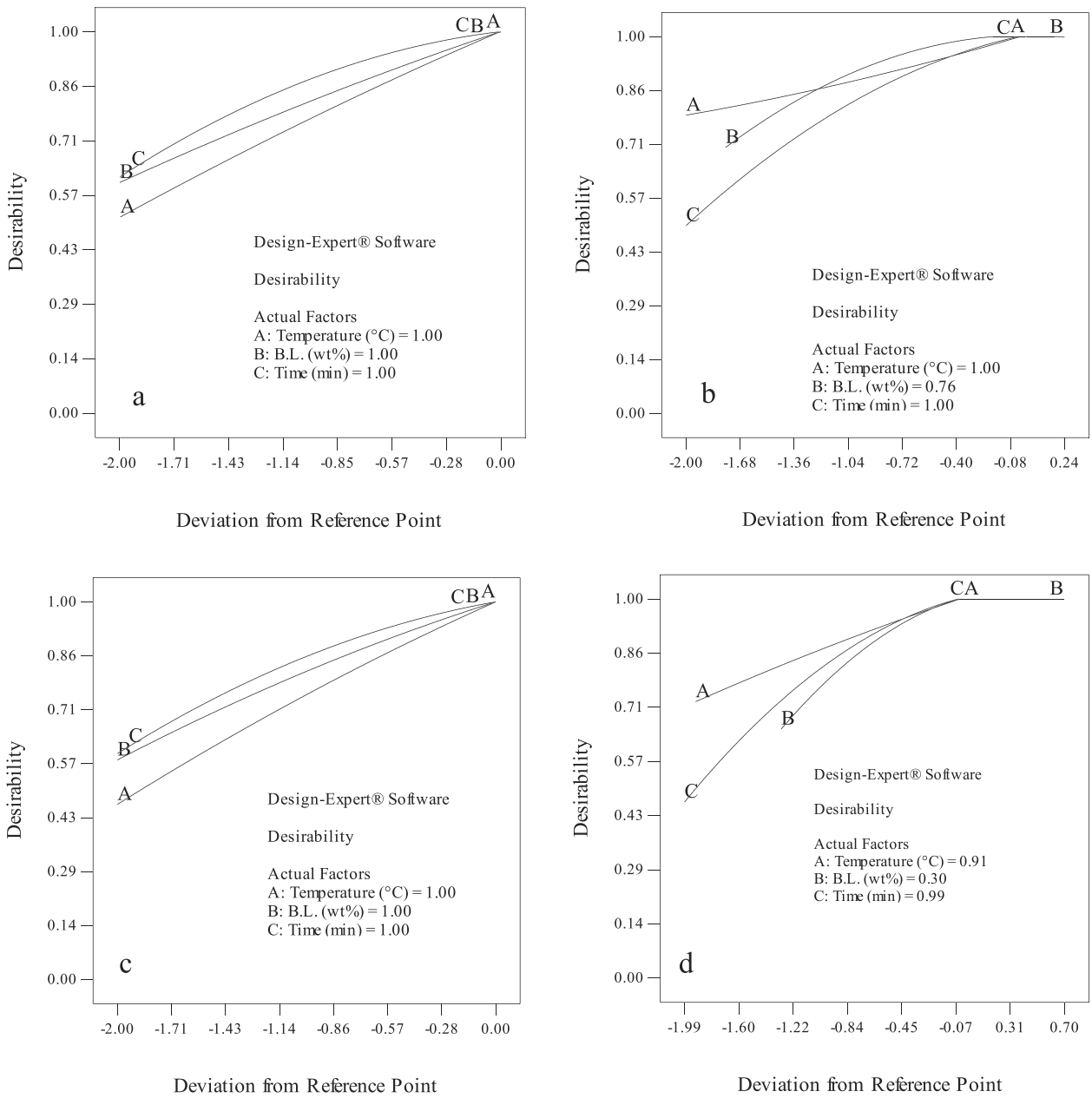

Fig. 5 – Perturbation plot for all the pretreatment experiments: (a) sugarcane bagasse, (b) bamboo by dilute sulfuric acid pretreatment; and (c) sugarcane bagasse, (d) bamboo by hot water pretreatment.

considering the 0.01 significance level. The Student's 't' distribution and corresponding values along with estimated parameter are given in Tables 5 and 6. The probability (P) values were used as a tool to check the significance of each coefficient.

Tables 5 and 6 shows the regression analysis for both the pretreatment experiments presenting the estimates and hypothesis tests to the regression coefficients. By applying the multiple regression analysis on experimental data, a second-order polynomial model representing TRS yield (gL^{-1}) in the experimental range studied with bamboo can be expressed by Equation (6) for hot water treatment and Equation (7) for dilute acid pretreatment, whereas hot water and dilute acid pretreatment of sugarcane bagasse was represented by Equation (8) and Equation (9) respectively.

$$TRS\ (gL^{-1}) = 14.82 + 1.27A + 1.63B + 2.46C - 0.10A^2 - 1.32B^2 - 0.93C^2 - 0.12AB + 0.23AC + 2.75BC \quad (6)$$

$$TRS\ (gL^{-1}) = 11.98 + 1.22X + 1.57Y + 2.47Z + 0.14X^2 - Y^2 - 0.84Z^2 - 0.21XY + 0.05XZ + 0.18YZ \quad (7)$$

$$TRS\ (gL^{-1}) = 17.93 + 3.63a + 2.75b + 2.56c - 0.39a^2 - 0.44b^2 - 0.98c^2 + 0.51\ ab + 0.55ac + 0.4bc \quad (8)$$

$$TRS\ (gL^{-1}) = 16.25 + 2.943x + 2.47y + 2.32z - 0.22x^2 - 0.23y2 - 1.01z^2 + 0.39xy + 0.42xz + 0.19yz \quad (9)$$

The quadratic models in Equations (6)–(9) with 10 terms contain 3 linear, 3 quadratic and 3 two-factor interactions terms.

Analysis of variance (ANOVA) was done by MINITAB-13. ANOVA (Tables 7 and 8) indicates that the effect of interactions of variables and their quadratic effects are highly significant and the regression between independent variables and output is also quite significant. The statistical significance of Equations (6)–(9) was checked by F-test and the ANOVA for the response surface quadratic model is shown in Tables 7 and 8 The results manifest that the model was highly significant as suggested by the model F value. The coefficient of variation (CV) specifies the degree of precision with which the treatments were compared. Higher the value of CV, lower the reliability of experiment. Here, lower CV values for dilute acid pre-treatment (0.23 and 0.12) and for hot water pre-treatment (3.1–3.16) indicates better precision and reliability of all the experiments.

The standardized residuals were also calculated to check the model adequacy. These residuals represent the difference between the response (actual) and predicted value that fitted best under the hypothesized model. In this study, the residual values were small which specified that the model prediction was accurate.

The precision of a model can be checked by R^2 value, normally a regression model having R^2 value higher than 0.9 is considered to have a very high correlation [17]. The closer the value to 1, better the correlation between the experimental and predicted values. Here, the value of R^2 (0.95) for Equation (7) indicates a close agreement between the experimental results and the theoretical values predicted by the model equation. Both linear and quadratic terms were significant at the 1% level. Therefore, the quadratic model was selected in this optimization study. The response surface plots are generated with one variable kept at its optimum level and varying the others within the experimental range (Figs. 3 and 4). The response surface plots can also described by the model equations (Equations (6)–(9)).

3.3. *Optimization of response and verification test*

The second-order polynomial model obtained in this study was utilized for response surface method optimization by using Design expert 7. A plateau region over which the production of TRS improved very little when time of operation crossed the "0" level with respect to both temperature and biomass loading (Fig. 3) irrespective of the type of biomass material. The production of TRS reached the maximum in the region of level "1" for both dilute acid and hot water pre-treatment of sugarcane bagasse and bamboo respectively. The numerical optimization process was conducted for each experiment, and the maximum amount of TRS was obtained using hot water pre-treatment with bagasse i.e. 26.51 gL^{-1} using all the parameters at level "1". The maximum yield of TRS for bamboo was obtained as 17.98 gL^{-1} using hot water pre-treatment method. The desired level (coded) for hot water process for pre-treatment of bamboo was as follows: temperature 0.92, biomass loading 0.32 and treatment time 1. The deviation from the reference point was not exceeding more than 0.7 for all the experiments (Fig. 5). The confirmatory experiments using the levels derived theoretically were conducted for hot water treatment process for both sugarcane bagasse and bamboo. The obtained experimental TRS yield for sugarcane bagasse and bamboo were 25.0 gL^{-1} and 17.5 gL^{-1} respectively.

4. Conclusions

The biomass selected for this study such as sugarcane bagasse and bamboo are abundant in the north east region of India. The physico-chemical characterization of biomass samples shows that these biomasses can act as the potential candidate for biofuel production in the north eastern region of India. The dilute acid pretreatment process is well known for delignification, apart from dilute acid pretreatment, present study also emphasized on hot water pretreatment. Both the pretreatment processes were thoroughly studied with respect to TRS yield (gL^{-1}) in the pretreated liquid. The phenomenon of both the pretreatment processes was explicated by RSM. The quadric equations facilitate to predict the yield of sugar at different combination of temperature, time and solid loadings respectively.

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REFERENCES

[1] Kralova I, Sjoblom J. Biofuels-renewable energy sources: a review. J Dispers Sci Technol 2010;31:409–25.
[2] McKendry P. Energy production from biomass (Part 1): overview of biomass. Bioresour Technol 2002;83:37–46.
[3] Varanasi P, Singh P, Auer P, Adams P, Simmons B, Singh S. Survey of renewable chemicals produced from lignocellulosic biomass during ionic liquid pretreatment. Biotechnol Biofuel 2013;6:14.
[4] Shill K, Padmanabhan S, Xin Q, Prausnitz JM, Clark DS, Blanch HW. Ionic liquid pretreatment of cellulosic biomass: enzymatic hydrolysis and ionic liquid recycle. Biotechnol Bioeng 2010;108:511–20.
[5] Sasmal S, Goud VV, Mohanty K. Ultrasound assisted Lime pretreatment of lignocellulosic biomass toward bioethanol production. Energy Fuels 2012;26:3777–84.
[6] Menon V, Rao M. Trends in bioconversion of lignocellulose: Biofuels, platform chemicals and biorefinery concept. Prog Energy Combust Sci 2012;38(4):522–50.
[7] Pessoa Jr A, Mancilha IM, Sato S. Acid hydrolysis of hemicellulose from sugarcane bagasse. Braz J Chem Eng 1997;14:10–3.
[8] Mauria S, Arora RK. Genetic resources of bamboos—an Indian perspective. Indian Forests 1988;114:539–48.
[9] Nath AJ, Das G, Das AK. Population structure and culm production of bamboos under traditional harvest regimes in Assam, Northeastern India. J Bamboo Rattan 2006;5(1–2):79–88.

[10] Source: Directorate of economics and Statistics, Department of Agriculture and Cooperation. Report 2009-2010.
[11] Naik S, Goud VV, Rout PK, Jacobson K, Dalai AK. Characterization of Canadian biomass for alternative renewable biofuel. Renew Energy 2010;35:1624–31.
[12] Lobovikov M, Paudel S, Piazza M, Ren H, Wu J. WORLD BAMBOO RESOURCES A thematic study prepared in the framework of the Global Forest Resources Assessment 2005. Food and Agriculture Organization of the United Nations Rome; 2007, ISBN 978-92-5-105781-0.
[13] Park S, Baker JO, Himmel ME, Parlla PA, Johnson DK. Cellulose crystallinity index: measurement techniques and their impact on interpreting cellulase performance. Biotechnol Biofuel 2010;3:10.
[14] Bansal P, Hall M, Realff MJ, Lee JH, Bommarus AS. Multivariate statistical analysis of X-ray data from cellulose: a new method to determine degree of crystallinity and predict hydrolysis rates. Bioresour Technol 2010;101:4461–71.
[15] Miller GL. Use of Dinitrosalicylic acid reagent for determination of reducing sugar. Anal Chem 1959;31:426–8.
[16] Segal L, Martin Jr Creely JJ, Canrad CM. An empirical method for estimating the degree of crystalinity of native cellulose using the X ray diffractometer. Text Res J 1959;29(10):786–94.
[17] Socrates G. Infrared and raman characteristic group frequencies: Tables and charts. 3rd ed. UK: Wiley Chichester; 2001.